

AeroLeaf/Wind Tree Technology- KSA Guide

1. What Is This Technology?

AeroLeaf / Wind Tree = Micro Renewable Energy Generator

It's a **biomorphic renewable energy solution** from French company **New World Wind**.

It looks like an artistic tree covered with *leaf-shaped mini wind turbines* called **Aeroleafs** that generate electricity from wind, and in hybrid models, solar too.

👉 Key tech features

- Each *leaf* is a micro wind turbine with a **vertical axis**, capturing wind from *all directions*.
- **Starts generating power at as low as ~2.5 m/s wind** (light breeze).
- Hybrid versions add **solar petals** that increase total energy output daily.

2. Products You Can Sell / Offer

Product	Description	Output (Normal)	Price (Indicative)
Aeroleaf	Single micro wind turbine "leaf"	~300 W	~€695 (+tax)
WindBush	Small 12-leaf tree	~3.6 kW	~€18,900 (+HT)
WindTree	Classic 36-leaf tree	~10.8 kW	~€50,990 (~5.4 kW nominal)
WindPalm	Modular tree (18–30 leaves)	~5.4–9 kW	~€28,800–€37,500
Hybrid Versions	Wind + solar petals	More annual output	+€300–€500 per unit

✓ All models can be roof-mounted, garden-installed, or used as public art & energy sources.

3. How Much Energy Can These Produce?

Example production data (typical conditions):

- A 36-leaf WindTree can produce up to **36,000 kWh/year** at 12 m/s wind speeds (~high wind).
- At more common winds (~8 m/s), output may be ~18,000 kWh/year.

➡ That's enough energy for **a typical small home + EV charging + lighting** depending on local consumption patterns.

➡ Combining hybrid solar petals increases total daily production.

4. Who Are Potential Customers?

◆ Residential Customers

- Homeowners looking to reduce electricity bills.
- Eco-conscious consumers.
- Owners of villas, coastal properties, green homes.

◆ Commercial / B2B

- Hotels / resorts looking for renewable credentials.
- Offices & co-working spaces.
- EV charging stations & community power hubs.

◆ Municipal / Institutional

- City parks & public spaces.
- Schools & college campuses.
- Urban beautification + clean energy deployment.

◆ Green Developers & ESG Projects

- Real estate projects claiming carbon neutrality.
 - CSR initiatives and sustainability milestones.
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. Business Model: How You Make Money

◆ Product Sales

Sell the devices directly (margin + markup).

Example: Buy WindTrees at €50,000, sell locally at market-adjusted price.

◆ Installation & Commissioning

Charge for installation, site survey, foundations, grid integration.

◆ Maintenance + Service Contracts

Annual service, cleaning, repair, firmware updates.

◆ Energy Fees

If allowed locally: sell generated energy to grid / neighbors.

◆ Leasing / Financing Packages

Offer monthly rental + long-term low payments (B2B especially).

◆ Carbon Credits & Government Incentives

Claim schemes if renewable energy certificates apply (depends on location).

6. Step-by-Step to Launch (Action Plan)

Step 1: Legal & Business Setup

- ✓ Register your company in local jurisdiction.
- ✓ Acquire GST/TIN, renewable energy licenses if needed.
- ✓ Trademark and branding (SolarTree / WindLeaf / etc).

Step 2: Partner with Manufacturer

- ✓ Contact *New World Wind / distributors* for bulk pricing or dealership.
- ✓ Negotiate import or exclusive rights if possible.

Step 3: Technical Training

- ✓ Learn installation, wiring, safety protocols.
- ✓ Train your tech team for maintenance.

Step 4: Supply Chain & Inventory

- ✓ Import initial units: WindTrees, Aeroleafs, hybrid kits.
- ✓ Stock small parts (controllers, mounts, screws).

Step 5: Site Survey & Pre-Sales

- ✓ Survey local wind patterns (wind speed maps or portable anemometers).
- ✓ Propose sites where consistent wind is available.

Step 6: Installation

- ✓ Provide groundwork/foundation.
- ✓ Connect to batteries / grid (inverters) depending on use.

Step 7: After Sales

- ✓ Provide remote monitoring, warranty, service plans.
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7. Cost & Investment Breakdown (Indicative)

Item	Estimated Cost (₹)
WindTree unit (36 Aeroleafs)	₹45–50 lakh
WindBush unit (12 Aeroleafs)	₹17–20 lakh
Single Aeroleaf (300W)	₹60,000–65,000 per leaf
Installation & civil works	₹50,000 – ₹2,00,000 per unit
BMS + Inverter + Battery system	₹1,00,000 – ₹5,00,000+ (depends on storage & load)
Shipping + Import duty + logistics	₹3–10 lakh (country, volume & port dependent)

Note: These are Europe averages. Local import duty and shipping to India will affect final landed costs.

8. Installation Logistics & Technical Tips

📌 Site Survey

- Measure average wind speeds (ideally >4–5 m/s).
- Avoid turbulence from buildings/trees blocking wind.

📌 Foundations / Mounting

- Use concrete base or rooftop mount kits.
- Ensure safe grounding and lightning protection.

📌 Energy Storage

- Combine with **batteries** to store peak generation.
- Use **hybrid solar** to balance day/night output.

📌 Grid / Off-grid Options

- For on-grid: integrate with local utility, net-meter where allowed.
- For off-grid: pair with solar + batteries for full autonomy.

9. Marketing & Sales Strategy

📌 Craft a Clear Value Proposition

- ➡ “Clean energy + stylish design + low noise + wildlife friendly.”
- ➡ “Reduce power bills + own power generation.”

📌 Channels

- Instagram, TikTok reels showing installation + real-world output.
- Partnerships with green builders / architects.
- Webinars on renewable energy for small businesses.

📌 Proof & Trust Signals

- Show **data on kWh generation** and **client testimonials**.
 - Highlight that they’re installed in cities worldwide.
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10. Risks & Things to Consider

⚠ Wind Variability

Wind speeds change seasonally, earnings may fluctuate.

⚠ Import & Taxes

Heavy import duties or regs may impact final price.

⚠ Maintenance

Although quiet, regular checks improve life span.

⚠ Realistic ROI

Understand typical ROI timelines: usually **5-10 years** depending on energy value.

📌 11. Starter Checklist

- ✓ Register business
- ✓ Partner with manufacturer
- ✓ Training + certification
- ✓ Initial stock (WindTrees, Bushes, Aeroleafs)
- ✓ Tools & installation crew
- ✓ Website + social proof
- ✓ Sales funnel + financing options
- ✓ After-sales + maintenance plan

